POST SECOND YEAR INDUSTRIAL TRAINING
{REPORT}

O7 Services IT Company

Submitted in partial fulfillment for the requirements for the degree of

Bachelor of Technology – COMPUTER SCIENCE WITH SPECIALISATION IN ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

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ABSTRACT

This report presents the work completed during a six-week industrial training program in the domain of Python and Data Science. The training was designed to provide practical exposure to data analysis techniques, Python programming, and the use of industry-relevant tools for extracting meaningful insights from real-world datasets. Throughout the training, emphasis was placed on developing analytical thinking, data preprocessing skills, visualization techniques, and project implementation.

As a part of the training, a project titled **Bank Fraud Exploratory Data Analysis (EDA)** was undertaken to analyse patterns and trends in banking transactions. The project involved cleaning and preprocessing the dataset, performing statistical analysis, identifying relationships among various features, and visualizing important patterns associated with fraudulent transactions. The objective was to gain meaningful insights into factors that may contribute to fraudulent activities and present the findings through interactive and informative visualizations.

The project was developed using Python and various data science libraries, including Pandas, NumPy, Matplotlib, Seaborn, Plotly, and Streamlit. Jupyter Notebook was used for data exploration and analysis, while Streamlit was utilized to build an interactive web-based dashboard for presenting the results.

The industrial training significantly enhanced practical knowledge of data analysis, visualization, and Python programming. It also provided valuable experience in working with real-world datasets, applying systematic analytical approaches, and effectively communicating insights through visual dashboards and technical documentation.

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Chapter 1: **INDUSTRIAL TRAINING**

1.1: Introduction

Industrial training is an integral part of professional education that bridges the gap between theoretical knowledge and practical application. It provides students with an opportunity to gain hands-on experience in a real-world working environment, enabling them to understand industry practices, workflows, and the technologies used by professionals. Such training enhances technical competence, problem-solving abilities, teamwork, communication skills, and overall professional development.

In today's technology-driven world, industries require professionals who possess not only strong academic knowledge but also practical skills and the ability to adapt to evolving technologies. Industrial training helps students become familiar with modern tools, programming languages, software development methodologies, and data-driven approaches that are widely used in the industry. It also encourages analytical thinking and enables students to apply classroom concepts to solve real-life problems.

As a part of the academic curriculum, I successfully completed a **six-week industrial training program** in the domain of **Python and Data Science**. The training focused on strengthening the fundamentals of Python programming, data manipulation, data visualization, and exploratory data analysis using widely adopted libraries and frameworks. Throughout the training, I gained practical exposure to working with datasets, performing data preprocessing, extracting meaningful insights, and presenting analytical results through interactive visualizations.

This industrial training played a significant role in enhancing my technical knowledge, analytical skills, and practical understanding of Python and Data Science. It has prepared me to approach real-world data analysis problems with greater confidence and has contributed to my overall professional and academic growth.

1.2: Company Profile

O7 Services is an ISO 9001:2015 Certified company founded in 2015. They specialize in a variety of services including Web Development, Mobile Application Development, Custom Software Development, UI/UX design, Hosting services, Digital Marketing, Registration of Domain Names with various extensions, AMC & MMC Services, Bulk SMS and voice calls. O7 Services offers advanced IT solutions supporting the entire business cycle - from consulting to system development, deployment, quality assurance, and 24x7 support. With over 9 years of experience, the company aims to form long-lasting strategic partnerships with clients, offering affordable prices, timely delivery, and measurable business results. Their headquarter is located in Jalandhar with a branch office in Hoshiarpur. Some of the products developed by O7 Services include Vehicle Tracking System, Invoice Software, School Management System, Hospital Management System, Parents - Teacher Communication App, Fee Management system, Task Management System, Online Food Ordering App, Security App, Admission system, Inventory Software, and Car Servicing App. Additionally, O7 Services provides training programs including 6 Weeks/6 Months Industrial Training, Project-Based Training, Corporate Training, and Job-Oriented Courses Training, covering major IT trends such as Full Stack Development (MEAN/MERN), Flutter, Kotlin Android, Swift UI (iOS), Firebase, Python, Angular, React.js, Vue.js, Node.js, ASP.NET, .NET Core, PHP, Laravel, CodeIgniter, Software Testing, Cloud Computing, Blockchain, DevOps, Data Science, Artificial Intelligence, Machine Learning, UI/UX Designing, Digital Marketing, WordPress, Linux, CCNP, CCNA Security, Network Security, Cyber Security, MCSE, MCITP, Java, Spring, Hibernate, C/C++, Photoshop, Adobe Illustrator, Figma, CorelDraw and many more.

1.3: Training Overview

The six-week industrial training program was conducted with the objective of providing practical knowledge and hands-on experience in the field of **Python and Data Science**. The training was designed to strengthen the understanding of Python programming while introducing various concepts and tools used in data analysis and visualization.
The training began with the fundamentals of Python, covering topics such as variables, data types, operators, control statements, functions, object-oriented programming, exception handling, and file handling. As the training progressed, emphasis was placed on data science concepts, including data collection, data preprocessing, data manipulation, exploratory data analysis (EDA), and data visualization.
Students were introduced to industry-standard Python libraries such as **Pandas** for data manipulation, **NumPy** for numerical computations, **Matplotlib** and **Seaborn** for static visualizations, **Plotly** for interactive charts, and **Streamlit** for developing interactive web-based dashboards. The training also included the use of **Jupyter Notebook**, which served as the primary environment for coding, experimentation, and data analysis.
A major component of the training involved working on a real-world project titled **"Bank Fraud Exploratory Data Analysis (EDA)."** The project required collecting insights from a banking transaction dataset through systematic data preprocessing, statistical analysis, and visualization techniques. Various analytical methods were applied to identify patterns, trends, and relationships within the data, enabling a deeper understanding of fraudulent transaction behaviour.
Throughout the training, equal importance was given to both theoretical concepts and practical implementation. Regular coding exercises, project development, and problem-solving activities helped in improving programming proficiency, analytical thinking, and the ability to work with real-world datasets. The training concluded with the successful completion and presentation of the project, demonstrating the practical application of the knowledge and skills acquired during the program.

1.4: Technologies Learned

During the six-week industrial training program, I gained practical knowledge of various programming languages, tools, libraries, and frameworks used in Python-based data analysis and visualization.

- **Python:** Used as the primary programming language for data analysisand project development.
- **Jupyter Notebook:** Used as the development environment for writing, executing, and documenting Python code. It facilitated interactive coding, data exploration, and visualization in an organized manner.

- **Pandas:** Utilized for data loading, cleaning, preprocessing, transformation, filtering, grouping, and statistical analysis. It served as the core library for handling structured datasets efficiently.
- **NumPy:** Used for numerical computations and efficient manipulation of multidimensional arrays. It provided support for mathematical and statistical operations required during data analysis.
- **Matplotlib:** Employed to create a variety of static visualizations such as bar charts, histograms, line graphs, scatter plots, and pie charts for analyzing dataset characteristics.
- **Seaborn:** Used to generate advanced statistical visualizations with enhanced aesthetics. It simplified the creation of heatmaps, box plots, count plots, violin plots, and correlation visualizations.
- **Plotly:** Used for developing interactive visualizations that allowed users to explore data dynamically through zooming, hovering, filtering, and other interactive features.
- **Streamlit:** Utilized to build an interactive web-based dashboard that presented the results of the Bank Fraud Exploratory Data Analysis project in a user-friendly and visually appealing manner.

Throughout the training, these technologies were applied collectively to perform data preprocessing, exploratory data analysis, visualization, and dashboard development. The practical implementation of these tools significantly enhanced my technical skills and provided valuable exposure to modern data science practices.

Chapter 2: **PROJECT OVERVIEW**

2.1: Introduction to the Project

The rapid growth of digital banking and online financial services has significantly improved the convenience and accessibility of financial transactions. However, this digital transformation has also led to an increase in fraudulent activities, making fraud detection a major challenge for financial institutions. Identifying suspicious transactions at an early stage is essential to minimize financial losses, protect customer information, and maintain trust in banking systems.

The project, **"Bank Fraud Exploratory Data Analysis (EDA)"**, was developed to analyse a banking transaction dataset and identify patterns, trends, and characteristics associated with fraudulent transactions. Rather than building a predictive machine learning model, the project focuses on understanding the data through systematic exploration and visualization. Exploratory Data Analysis helps uncover hidden relationships, detect anomalies, and generate valuable insights that can support future fraud detection strategies.

The project involved several stages, including data loading, preprocessing, cleaning, handling missing values, analyzing feature distributions, and studying the relationships between different variables. Various statistical techniques and visualization methods were used to examine customer demographics, transaction characteristics, payment methods, merchant categories, transaction timings, authentication attempts, and geographical information in relation to fraudulent activities.

The primary goal of this project was to transform raw banking transaction data into meaningful information that highlights potential fraud patterns and supports informed decision-making. By applying Exploratory Data Analysis techniques, the project demonstrates how data-driven insights can contribute to understanding fraudulent behaviour and assist financial institutions in strengthening fraud monitoring and prevention efforts.

2.2: Problem Statement

Fraud has become one of the most significant challenges faced by the banking and financial sector. As the volume of financial transactions continues to grow across various banking channels, detecting fraudulent activities has become increasingly complex. Fraudulent transactions can occur through different payment methods, merchant categories, transaction types, and geographical locations, making it difficult to identify suspicious patterns through manual inspection alone. Banking institutions generate large amounts of transaction data every day. This data contains valuable information that can help uncover unusual behaviours, transaction trends, and risk factors associated with fraud. However, without systematic analysis, these hidden patterns often remain unnoticed, limiting the ability to understand the characteristics of fraudulent activities. The problem addressed in this project is to perform Exploratory Data Analysis (EDA) on a banking transaction dataset to identify patterns, trends, and anomalies related to fraudulent transactions. The project focuses on examining various transaction attributes, including customer demographics, transaction details, payment methods, merchant categories, authentication attempts, transaction timing, and geographical information, to understand how these factors are associated with fraud.

The insights obtained from this analysis can support financial institutions in recognizing high-risk transaction characteristics, improving fraud monitoring practices, and strengthening decision-making processes. By converting raw transaction data into meaningful visualizations and analytical insights, the project demonstrates the importance of data-driven analysis in understanding and addressing banking fraud.

2.3: Objectives of the Project

The primary objective of this project is to perform **Exploratory Data Analysis** on a banking transaction dataset to gain meaningful insights into the characteristics and patterns associated with fraudulent transactions. The project aims to transform raw data into useful information through systematic analysis and visualization.
The specific objectives of the project are as follows:

1. To understand the structure and characteristics of the banking transaction dataset.
2. To clean and preprocess the data by handling missing values, removing inconsistencies, and preparing it for analysis.
3. To perform exploratory data analysis using statistical methods and visualizations to identify patterns and trends.
4. To analyse the relationship between transaction attributes and fraudulent activities.
5. To examine the impact of factors such as transaction amount, payment method, merchant category, transaction timing, geographical location, authentication attempts, and customer demographics on fraud occurrence.
6. To identify high-risk transaction characteristics and highlight potential indicators of fraudulent behaviour.
7. To uncover hidden patterns, trends, and characteristics associated with fraudulent activities.
8. To present analytical findings through clear, informative, and interactive visualizations.
9. To develop a user-friendly dashboard using **Streamlit** for effective presentation and exploration of the analysis results.
10. To demonstrate the practical application of Python and data science libraries in solving real-world analytical problems.
11. To provide insights that can support better fraud monitoring, risk assessment, and decision-making in the banking sector.

2.4: Scope of the Project

The scope of this project is centered on the **exploratory analysis of banking transaction data** to identify patterns, trends, and characteristics associated with fraudulent activities. The project focuses on understanding the dataset through data preprocessing, statistical analysis, and visualization, enabling meaningful insights to be extracted from large volumes of transaction records.
The analysis includes examining various aspects of the dataset, such as customer demographics, transaction details, payment methods, merchant categories, transaction timing, authentication attempts, device information, and geographical attributes. Different visualization techniques are used to study the distribution of these features and their relationship with fraudulent transactions.
The project also includes the development of an interactive dashboard using **Streamlit**, allowing users to explore the analytical results through dynamic charts and visualizations. This enhances the accessibility and presentation of insights, making the analysis more intuitive and user-friendly.
However, the scope of the project is limited to **Exploratory Data Analysis (EDA)** and does not include the development or deployment of predictive machine learning models for fraud detection. The insights generated are intended to support a better understanding of fraud patterns and assist in identifying areas that may require closer monitoring.

Overall, the project demonstrates how data analysis and visualization techniques can be applied to banking transaction data to uncover valuable information that can aid fraud monitoring, improve analytical decision-making, and serve as a foundation for future research and advanced fraud detection systems.

2.5: Technology Stack

The **Bank Fraud Exploratory Data Analysis (EDA)** project was developed using a combination of programming languages, libraries, development environments, and visualization tools. Each technology was selected based on its capabilities in data processing, analysis, visualization, and dashboard development.

2.5.1 Programming Language

- **Python**
  1. Used as the primary programming language for the entire project.
  2. Chosen for its simplicity, extensive library ecosystem, and strong support for data science applications.

2.5.2 Development Environment

- **Jupyter Notebook**
  1. Used for writing and executing Python code.
  2. Enabled interactive data exploration and documentation.
  3. Facilitated experimentation with preprocessing techniques and visualizations.

2.5.3 Exploratory Data Analysis (EDA)

- **Pandas**
  1. Data loading and manipulation.
  2. Data cleaning and preprocessing.
  3. Handling missing values and duplicate records.
- **NumPy**
  1. Numerical computations.
  2. Mathematical and statistical calculations.

2.5.4 Static Data Visualization

- **Matplotlib**
  1. Creation of fundamental charts and graphs.
  2. Histograms, bar charts, line plots, scatter plots, and pie charts.
  3. Visualization of feature distributions and trends.
- **Seaborn**
  1. Count plots, box plots, violin plots, heatmaps.
  2. Enhanced aesthetics for analytical charts.
  3. Correlation analysis and comparative visualizations.

2.5.5 Interactive Data Visualization

- **Plotly**
  1. Development of interactive charts.
  2. Hover information, zooming, filtering, and dynamic exploration.
  3. Used to improve user interaction and data interpretation within the dashboard.

2.5.6 Dashboard Development

- **Streamlit**
  1. Framework used to develop the interactive web application.
  2. Displayed analytical results through an organized and user-friendly interface.
  3. Integrated interactive charts, metrics, dataset exploration, and project insights.

2.5.7 User Interface Styling

- **HTML**
  1. Used to structure custom components and content within the Streamlit application where required.
- **CSS**
  1. Applied to customize the appearance of the dashboard.
  2. Improved layout, typography, spacing, colors, buttons, cards, and overall user experience beyond Streamlit's default styling.

2.5.8 Version Control

- **Git & GitHub**
  1. Used for maintaining the project repository and managing source code.
  2. Facilitated version control, project organization, and code sharing.

Chapter 3: **PROJECT DEVELOPMENT**

3.1: Dataset Description

The dataset used in this project consists of **1,000,000 banking transaction records**, representing a wide range of customer and transaction information. It was selected to perform an in-depth Exploratory Data Analysis (EDA) for understanding the characteristics and patterns associated with fraudulent transactions. The dataset contains **26 attributes**, covering transaction details, customer information, account characteristics, geographical data, authentication-related features, and fraud indicators.

The dataset occupies approximately **653.69 MB** of memory and contains a combination of numerical and categorical variables. Out of the 26 attributes, **11 are integer-type variables**, **5 are floating-point variables**, and **10 are categorical (object) variables**. Most columns contain complete information with no missing values. The only exception is the **fraud_type** column, which contains values only for transactions identified as fraudulent, while non-fraudulent transactions have null values in this field.

The dataset includes the following categories of information:

1. **Transaction Information:** Transaction ID, transaction date, transaction time, transaction amount, hour of transaction, transaction frequency, and time elapsed since the previous transaction.
2. **Customer Information:** Customer ID, customer age, and credit score.
3. **Account Information:** Account age and account balance.
4. **Location Information:** Country, city, distance from home, and whether the transaction was international.
5. **Transaction Characteristics:** Merchant category, payment method, and device type used for the transaction.
6. **Security Features:** Number of failed authentication attempts and whether the customer's PIN was recently changed.
7. **Fraud Indicators:** A binary target variable (**is_fraud**) indicating whether a transaction is fraudulent, along with the corresponding **fraud_type** for fraudulent transactions.

The numerical attributes cover a broad range of values. For example, customer ages range from **18 to 85 years**, credit scores range from **300 to 850**, account balances vary from **100 to 500,000**, and transaction amounts range from **1.00 to 46,129.60**. These diverse values make the dataset suitable for identifying patterns across different customer profiles and transaction behaviours.

Overall, the dataset provides a comprehensive representation of banking transactions by combining customer demographics, financial information, transaction characteristics, geographical details, and security-related features. This diversity enables detailed exploratory analysis and facilitates the extraction of meaningful insights into factors associated with fraudulent activities.

transaction_id	customer_id	transaction_	transactic	hour_o	is_wee	is_nigh	country	city	merchant_catego	payment_meth	device_type
TXN0000000001	CUST00121959	2023-08-17	21:13:00	21	0	0	USA	London	Grocery	Bank Transfer	POS Terminal
TXN0000000002	CUST00146868	2024-02-06	05:16:00	5	0	1	UK	New York	Healthcare	Cheque	Desktop
TXN0000000003	CUST00131933	2024-06-28	12:15:00	12	0	0	Canada	Delhi	Grocery	Crypto	Mobile
TXN0000000004	CUST00103695	2023-03-16	02:53:00	2	0	1	France	Tokyo	Utilities	Debit Card	Mobile
TXN0000000005	CUST00119880	2024-07-12	12:39:00	12	0	0	Canada	Melbourne	Clothing	Debit Card	Desktop
TXN0000000006	CUST00110269	2024-06-26	04:28:00	4	0	1	Canada	Guadalajara	Crypto Exchange	Debit Card	Mobile
TXN0000000007	CUST00054887	2020-09-05	19:22:00	19	1	0	USA	Mumbai	Education	Debit Card	Mobile
TXN0000000008	CUST00137338	2024-09-02	14:02:00	14	0	0	UK	Melbourne	Grocery	Crypto	POS Terminal
TXN0000000009	CUST00168267	2021-12-07	03:23:00	3	0	1	USA	Melbourne	Online Shopping	Debit Card	Mobile
TXN0000000010	CUST00087499	2020-03-30	21:08:00	21	0	0	Mexico	Mumbai	Crypto Exchange	Credit Card	Mobile

transaction_id	custom	credit_	accoun	account_t	transact	num_p	transac	distanc	time_si	is_inter	failed_	pin_ch	is_frau	fraud_type
TXN0000000001	18	695	7.2	369.36	39.49	157	23	52.7	10.2	0	0	0	0	None
TXN0000000002	30	600	8.6	2681.99	153.71	153	23	0.9	12.47	0	0	0	0	None
TXN0000000003	20	711	10.8	44590.64	118.2	161	20	9.2	0.08	0	1	0	0	None
TXN0000000004	29	711	7.3	1213.37	49.5	160	25	14.8	17.94	1	0	1	1	Synthetic Id
TXN0000000005	49	751	11.8	34615.01	30.74	134	18	38.9	2.16	0	0	0	0	None
TXN0000000006	22	681	3.7	11181.08	35.36	157	19	23	31.46	0	2	0	1	Friendly Fra
TXN0000000007	56	733	4.6	11202.02	92.53	178	14	5	7.5	1	0	1	0	None
TXN0000000008	47	658	0.9	19545.95	23.12	145	19	6	6.4	0	0	0	0	None
TXN0000000009	48	656	4.5	1061.08	330.47	164	23	27.8	47.27	0	0	0	0	None
TXN0000000010	70	836	0.7	7888.36	139.81	154	29	1.2	16.54	0	0	0	0	None

Figure 3.1: Raw Dataset Overview

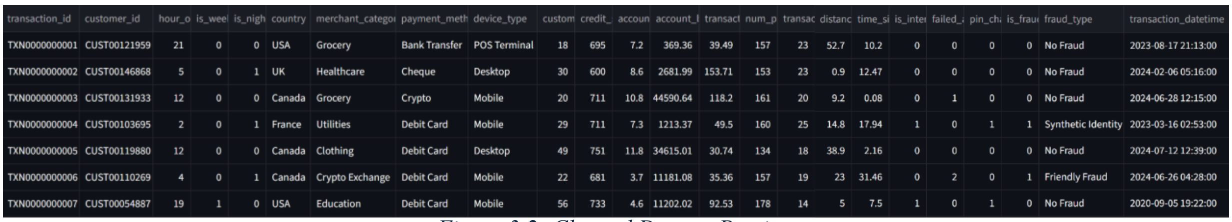
3.2: Data Preprocessing

Before performing Exploratory Data Analysis (EDA), the dataset was thoroughly examined and preprocessed to ensure its quality, consistency, and suitability for analysis. Data preprocessing is a crucial step in any data analysis workflow, as it improves data reliability and helps produce accurate analytical results.

The preprocessing steps carried out in this project are summarized below:

1. **Duplicate Record Analysis:** The dataset was checked for duplicate records to avoid redundant information that could affect the analysis. The examination revealed that **no duplicate entries** were present, indicating that each transaction record was unique.
2. **Missing Value Handling:** The dataset was inspected for missing values across all attributes. It was observed that only the **fraud_type** column contained null values. Further analysis showed that these missing values corresponded exclusively to transactions where the **is_fraud** attribute had a value of **0**, indicating non-fraudulent transactions. Therefore, these null values were replaced with **"No Fraud"** to improve data consistency and simplify further analysis.
3. **Feature Selection:** The **city** attribute was removed from the dataset because it was not utilized in the current analysis and did not contribute to the objectives of the project. Removing unnecessary features helped simplify the dataset and reduced memory usage.
4. **Date and Time Transformation:** The **transaction_date** and **transaction_time** columns were merged to create a single **transaction_datetime** attribute. The newly created column was then converted to the **datetime** data type, enabling efficient time-based operations and improving compatibility with date and time analysis.
5. **Memory Optimization:** To improve computational efficiency, the dataset was optimized after preprocessing. Through feature refinement and appropriate data handling, the memory consumption was reduced from **653.69 MB** to **576.17 MB**, resulting in a more efficient dataset for analysis and visualization.

After completing these preprocessing steps, the dataset was clean, consistent, and well-structured for Exploratory Data Analysis. The processed dataset served as the foundation for identifying fraud-related patterns, generating visualizations, and developing the interactive dashboard.



transaction_id	customer_id	hour_o	is_weel	is_nigh	country	merchant_categoı	payment_meth	device_type	custom	credit_	accoun
TXN0000000001	CUST00121959	21	0	0	USA	Grocery	Bank Transfer	POS Terminal	18	695	7.2
TXN0000000002	CUST00146868	5	0	1	UK	Healthcare	Cheque	Desktop	30	600	8.6
TXN0000000003	CUST00131933	12	0	0	Canada	Grocery	Crypto	Mobile	20	711	10.8
TXN0000000004	CUST00103695	2	0	1	France	Utilities	Debit Card	Mobile	29	711	7.3
TXN0000000005	CUST00119880	12	0	0	Canada	Clothing	Debit Card	Desktop	49	751	11.8
TXN0000000006	CUST00110269	4	0	1	Canada	Crypto Exchange	Debit Card	Mobile	22	681	3.7
TXN0000000007	CUST00054887	19	1	0	USA	Education	Debit Card	Mobile	56	733	4.6

transaction_id	account_l	transact	num_p	transac	distanc	time_si	is_inte	failed_	pin_ch	is_frau	fraud_type	transaction_datetime
TXN0000000001	369.36	39.49	157	23	52.7	10.2	0	0	0	0	No Fraud	2023-08-17 21:13:00
TXN0000000002	2681.99	153.71	153	23	0.9	12.47	0	0	0	0	No Fraud	2024-02-06 05:16:00
TXN0000000003	44590.64	118.2	161	20	9.2	0.08	0	1	0	0	No Fraud	2024-06-28 12:15:00
TXN0000000004	1213.37	49.5	160	25	14.8	17.94	1	0	1	1	Synthetic Identity	2023-03-16 02:53:00
TXN0000000005	34615.01	30.74	134	18	38.9	2.16	0	0	0	0	No Fraud	2024-07-12 12:39:00
TXN0000000006	11181.08	35.36	157	19	23	31.46	0	2	0	1	Friendly Fraud	2024-06-26 04:28:00
TXN0000000007	11202.02	92.53	178	14	5	7.5	1	0	1	0	No Fraud	2020-09-05 19:22:00

Figure 3.2: Cleaned Dataset Preview

3.3: Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a fundamental step in the data analysis process that involves examining, summarizing, and visualizing a dataset to understand its underlying structure and characteristics. The primary objective of EDA is to identify patterns, trends, relationships, and anomalies within the data before drawing conclusions or applying advanced analytical techniques. It enables analysts to gain a deeper understanding of the dataset and supports informed decision-making based on evidence derived from the data.

In this project, Exploratory Data Analysis was performed on a banking transaction dataset to investigate the characteristics of fraudulent and non-fraudulent transactions. A systematic approach was followed to explore the dataset from multiple perspectives, including customer demographics, account information, transaction attributes, merchant categories, payment methods, geographical factors, transaction timing, and security-related features.

The analysis began with an examination of individual variables to understand their distributions and statistical properties. This was followed by studying the relationships between different features and the fraud indicator to identify factors that exhibited significant variations between fraudulent and legitimate transactions. Correlation analysis was also performed to evaluate the degree of association among numerical variables.

A wide range of visualization techniques was employed to present the findings in an intuitive and informative manner. Histograms, count plots, bar charts, box plots, scatter plots, pie charts, heatmaps, and interactive visualizations were created using **Matplotlib**, and **Seaborn**. These visualizations facilitated the interpretation of complex data patterns and enabled effective comparison across different transaction characteristics.

To organize the analysis effectively, the Exploratory Data Analysis was divided into three major stages:

1. **Univariate Analysis:** Examination of individual features to understand their distribution, frequency, and statistical characteristics.
2. **Bivariate Analysis:** Investigation of relationships between individual features and the fraud indicator to identify potential fraud-associated patterns.
3. **Multivariate Analysis:** Exploration of relationships involving multiple variables simultaneously to gain deeper insights into transaction behaviour and fraud characteristics.

The insights obtained through Exploratory Data Analysis formed the foundation of the project and were subsequently incorporated into an interactive dashboard, enabling users to explore the analytical results through an intuitive and user-friendly interface.

3.4: Univariate Analysis

Univariate Analysis involves the examination of individual variables independently to understand their distribution, frequency, central tendency, variability, and overall characteristics. It serves as the first stage of Exploratory Data Analysis (EDA), providing valuable insights into the behaviour of each feature without considering its relationship with other variables. This analysis helps identify patterns, detect anomalies, and gain a comprehensive understanding of the dataset's composition.

In this project, univariate analysis was performed on both numerical and categorical features of the banking transaction dataset using descriptive statistics and various visualization techniques, including histograms, count plots, bar charts, box plots, and pie charts. The objective was to examine the distribution of transaction attributes, customer characteristics, account information, and fraud-related features individually, thereby establishing a foundation for the subsequent bivariate and multivariate analyses.

3.4.1: Distribution of Transaction Amounts

The distribution of transaction amounts indicates that the majority of transactions are of relatively low value, while only a small proportion involve high-value amounts. These high-value transactions appear as outliers and increase the overall average transaction amount, causing the mean (204.79) to be considerably higher than the median (73.12). This suggests that the dataset is dominated by smaller transactions with a limited number of exceptionally large transactions.

3.4.2: Distribution of Account Balances
The distribution of customer account balances shows that most customers maintain relatively low to moderate account balances, whereas only a small proportion possess significantly higher balances. The presence of these high-balance accounts increases the average account balance (16,594.25), making it substantially higher than the median (8,092.12). This indicates considerable variation in customer account balances across the dataset.

3.4.3: Distribution of Customer Ages
The customer age distribution ranges from **18 to 85 years**, representing an adult banking population. The mean age (41.77 years) is nearly equal to the median (42 years), indicating a fairly balanced distribution. Most customers fall within the 32 to 51 years age group, and no unrealistic or anomalous age values are observed, suggesting that the dataset contains valid and consistent customer age information.

3.4.4: Distribution of Customer Credit Scores
The customer credit scores range from **300 to 850**, covering the full spectrum of creditworthiness. The mean credit score (**679.03**) is almost identical to the median (**680**), indicating a fairly balanced distribution. Most customers have credit scores between **625 and 734**, suggesting that the majority fall within the moderate to good credit score range. Additionally, no unrealistic or anomalous credit score values are observed, indicating good data quality.

3.4.5: Distribution of Customer Account Ages
The account age ranges from **0.1 to 30 years**, representing a diverse mix of newly opened and long-established customer accounts. The mean account age (**4.99 years**) is slightly higher than the median (**3.5 years**), indicating that a relatively small number of older accounts increase the overall average. Most customer accounts have been active for **1.4 to 6.9 years**, reflecting a predominantly young to moderately established customer base. No unrealistic account age values are observed in the dataset.

3.4.6: Distribution of Transactions by Hour of Day
The hourly distribution of transactions is relatively uniform, with transaction counts remaining consistent throughout the 24-hour period. The number of transactions per hour varies only slightly, ranging from approximately **41,100 to 41,900**, indicating that transaction activity is evenly spread across the day rather than concentrated during specific hours. No significant spikes, drops, or unusual patterns are observed in the hourly transaction distribution.

3.4.7: Distribution of Distance from Home
The distance between customers' homes and transaction locations ranges from **0.0 km to 292.1 km**, indicating considerable variation in transaction locations. The mean distance (**20.01 km**) is higher than the median (**13.9 km**), reflecting the presence of a relatively small number of transactions occurring at much greater distances. Most transactions take place within **5.8 km to 27.7 km** of the customer's home, while a few long-distance transactions appear as outliers in the dataset.

3.4.8: Distribution of Time Since Previous Transaction

The time elapsed since the previous transaction ranges from **0.0 to 154.2 hours**, indicating substantial variation in customer transaction intervals. The mean interval (**12.00 hours**) exceeds the median (**8.32 hours**), suggesting that a relatively small number of transactions occur after much longer time gaps. Most transactions take place within **3.45 to 16.63 hours** of the previous transaction, while a few transactions with exceptionally long intervals appear as outliers in the dataset.

3.4.9: Distribution of Number of Previous Transactions

The number of previous transactions ranges from **96 to 213**, reflecting moderate variation in customers' transaction histories. The mean (**150.00**) is nearly identical to the median (**150**), indicating a fairly balanced distribution. Most customers have completed between **142 and 158** previous transactions, suggesting a relatively consistent level of historical transaction activity across the dataset. No significant anomalies or extreme values are observed.

3.4.10: Distribution of Monthly Transaction Frequency

The monthly transaction frequency ranges from **2 to 44 transactions**, indicating moderate variation in customer transaction activity. The mean monthly frequency (**19.99**) is almost identical to the median (**20**), reflecting a fairly balanced distribution. Most customers perform between **17 and 23 transactions per month**, suggesting a consistent level of monthly banking activity. No significant skewness or unusual transaction frequency values are observed in the dataset.

3.4.11: Distribution of Merchant Categories

The dataset comprises transactions across **15 distinct merchant categories**, representing a broad range of customer spending activities. Transaction counts are distributed fairly evenly among all categories, with each contributing approximately **66,000 to 67,000 transactions**. Although **Entertainment**, **Jewelry**, and **Electronics** record slightly higher transaction volumes, the differences are minimal. Overall, no single merchant category dominates the dataset, indicating a well-balanced representation of transaction types.

3.4.12: Distribution of Payment Methods

The dataset includes **six distinct payment methods**, representing a variety of transaction channels. **Credit Card** is the most commonly used payment method, followed by **Debit Card**, indicating a strong preference for card-based transactions. **Bank Transfer** and **Mobile Payment** show moderate usage, while **Crypto** and **Cheque** account for the lowest number of transactions. Overall, the distribution reflects varying customer preferences across different payment methods.

3.4.13: Distribution of Device Types

The dataset contains transactions originating from **five distinct device types**, representing multiple banking and transaction platforms. **Mobile devices** account for the highest number of transactions, followed by **Desktop** devices. **ATM**, **Tablet**, and **POS Terminal** each contribute a smaller but relatively similar share of transactions. Overall, the distribution indicates that customers predominantly perform transactions using mobile devices, highlighting the widespread adoption of mobile banking and digital transaction platforms.

3.4.14: Distribution of Transactions by Day Type

The distribution of transactions by day type shows that approximately **71.4%** of all transactions occur on **weekdays**, while the remaining **28.6%** take place on **weekends**. This indicates that customer transaction activity is considerably higher during the working week, with weekdays accounting for the majority of banking transactions.

3.4.15: Distribution of Daytime and Nighttime Transactions

Approximately **62.5%** of all transactions occur during the **daytime**, while **37.5%** take place during **nighttime** hours. This indicates that customer transaction activity is generally higher during the day. However, the considerable proportion of nighttime transactions suggests that banking transactions are carried out throughout the day rather than being limited to standard business hours.

3.4.16: Distribution of Domestic and International Transactions

Approximately **85.0%** of all transactions are **domestic**, while **15.0%** are **international**. This indicates that the majority of customer transactions are conducted within the home country. Although international transactions account for a smaller share of the dataset, they still represent a meaningful proportion of overall transaction activity.

3.4.17: Distribution of Recent PIN Changes

Approximately **92.0%** of customers have **not** changed their PIN recently, while only **8.0%** have performed a recent PIN change. This indicates that recent PIN updates are relatively infrequent, with the vast majority of customers retaining the same PIN during the observed period.

3.4.18: Distribution of Fraudulent and Legitimate Transactions

Approximately **94.5%** of the transactions are classified as **legitimate**, while **5.53%** are identified as **fraudulent**. This indicates a clear class imbalance, with legitimate transactions forming the overwhelming majority of the dataset. Despite representing a relatively small proportion, the dataset contains **more than 55,000 fraudulent transactions**, providing a substantial number of fraud cases for meaningful exploratory analysis.

3.4.19: Distribution of Failed Authentication Attempts

The distribution of failed authentication attempts shows that approximately **80.0%** of transactions are completed without any failed attempts. As the number of failed attempts increases, the frequency of transactions decreases steadily, indicating that multiple authentication failures are relatively uncommon. Transactions involving **four or five failed attempts** represent only a small proportion of the dataset, resulting in a distribution that is heavily concentrated at zero failed attempts.

3.4.20: Distribution of Fraud Types

Fraudulent transactions in the dataset are classified into **six distinct fraud types**, representing a diverse range of fraudulent activities. **Phishing** is the most frequently occurring fraud type, followed closely by **Account Takeover**, **Synthetic Identity**, **Card Cloning**, **Friendly Fraud**, and **Identity Theft**. The number of transactions across all fraud categories is relatively similar, with each fraud type contributing approximately **9,200 fraudulent transactions**. This balanced distribution ensures that no single fraud type disproportionately influences the analysis.

3.4.21: Distribution of Transactions by Country

The dataset contains transactions from **ten different countries**, representing a geographically diverse customer base. The **United States (USA)** accounts for the highest number of transactions, followed by **India** and the **United Kingdom (UK)**. The remaining transactions are distributed across **Brazil, Germany, Japan, France, Canada, Mexico,** and **Australia**. Overall, the distribution is uneven, with the USA contributing a substantially larger share of transactions than the other countries represented in the dataset.

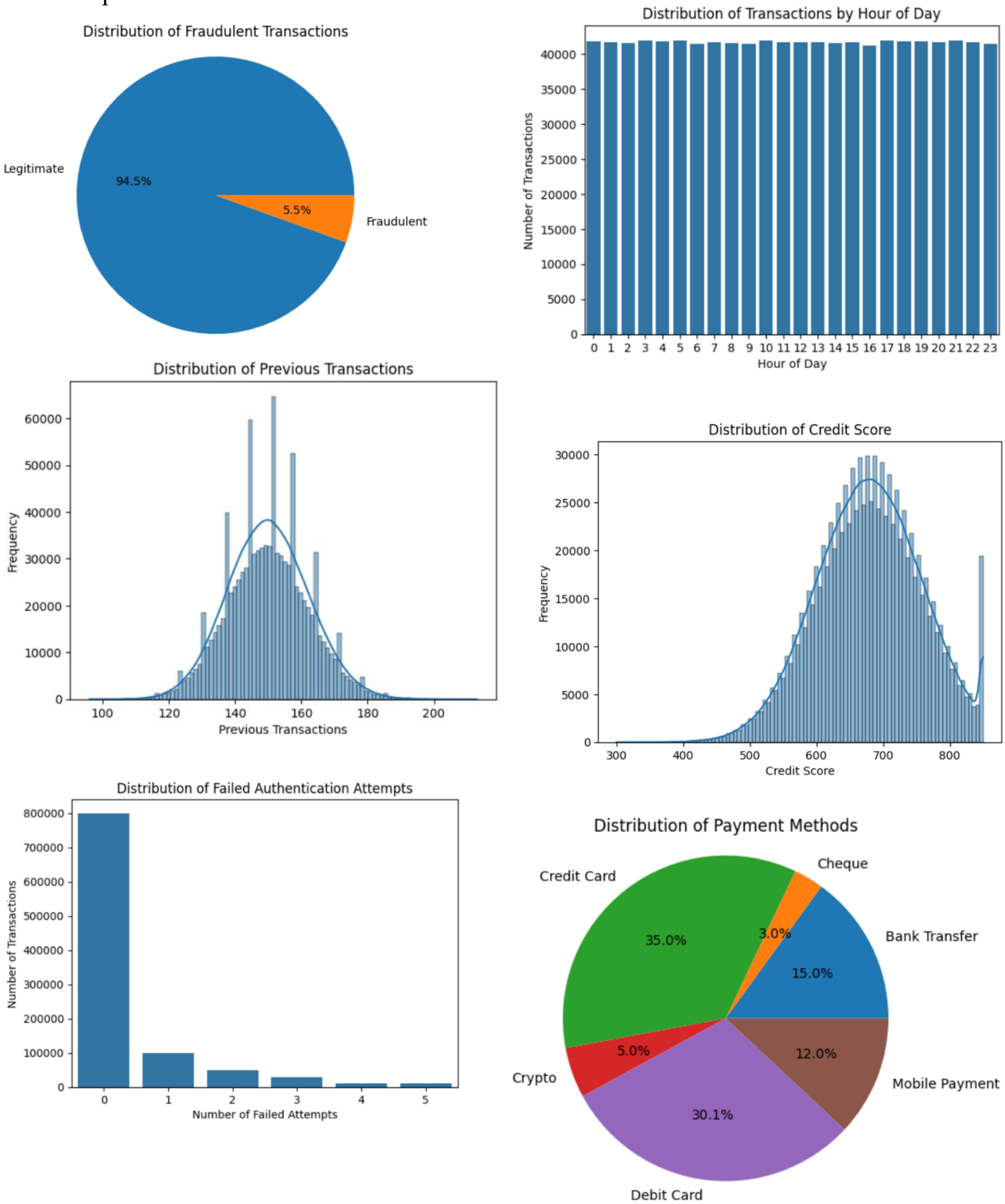


Figure 3.3: Few Univariate Distributions

3.5 Bivariate Analysis

Bivariate Analysis involves the simultaneous examination of two variables to identify relationships, associations, and patterns between them. In the context of Exploratory Data Analysis (EDA), it helps determine how individual features influence or relate to a target variable, enabling a deeper understanding of the factors associated with a particular outcome.
In this project, bivariate analysis was performed to examine the relationship between various transaction, customer, account, geographical, and security-related features and the **fraud indicator (is_fraud)**. Comparative analysis and visualizations were used to identify differences in fraud occurrence across different categories and numerical variables. Techniques such as grouped bar charts, box plots, count plots, and statistical summaries were employed to highlight meaningful relationships and reveal characteristics commonly associated with fraudulent transactions.
The observations obtained from this analysis provide valuable insights into how different transaction attributes vary between legitimate and fraudulent transactions, forming the basis for identifying potential fraud indicators and supporting more informed decision-making in the banking domain.

3.5.1: Transaction Amounts by Fraud Status

Fraudulent transactions exhibit a slightly higher transaction amount than legitimate transactions, with both the median (**76.10** vs. **72.95**) and the mean (**228.08** vs. **203.36**) being greater for fraudulent transactions. Although fraudulent transactions show higher variability in transaction amounts, as indicated by a larger standard deviation, both groups display similar right-skewed distributions with several high-value outliers. The substantial overlap in their interquartile ranges suggests that **transaction amount alone is not a strong indicator for distinguishing fraudulent transactions from legitimate ones**.

3.5.2: Account Balance by Fraud Status

Customers involved in fraudulent transactions have a marginally higher account balance than those involved in legitimate transactions, with slight increases in both the median (**8,166.15** vs. **8,087.67**) and the mean (**16,791.77** vs. **16,582.70**). Both groups exhibit similar right-skewed distributions with a small number of exceptionally high account balances. The considerable overlap between the two distributions indicates that **account balance alone provides limited ability to distinguish fraudulent transactions from legitimate ones**.

3.5.3: Customer Age by Fraud Status

The age distributions of customers involved in legitimate and fraudulent transactions are highly similar. The median age differs only slightly (**42 years** for legitimate transactions and **41 years** for fraudulent transactions), while the mean ages are also nearly identical (**41.78** and **41.61 years**, respectively). The substantial overlap between the two distributions indicates that **fraudulent transactions occur across a wide range of customer ages**, suggesting that customer age alone is not a significant distinguishing factor for fraud.

3.5.4: Credit Score by Fraud Status

The credit score distributions for legitimate and fraudulent transactions are highly comparable. The median credit scores (**680** for legitimate transactions and **678** for fraudulent transactions) and the mean credit scores (**679.13** and **677.33**, respectively) differ only marginally. The substantial overlap between the two distributions indicates that **fraudulent transactions occur across a broad range of credit scores**, suggesting that credit score alone is not a strong differentiating factor for fraudulent activity.

3.5.5: Account Age by Fraud Status

The account age distributions for legitimate and fraudulent transactions are virtually identical. Both groups have a median account age of **3.5 years** and an average account age of approximately **4.99 years**, with similar variability and identical interquartile ranges (**1.4 to 6.9 years**). The extensive overlap between the two distributions indicates that **fraudulent transactions occur across both newly opened and long-established accounts**, suggesting that account age alone is not a reliable indicator of fraud.

3.5.6: Transaction Hour by Fraud Status

Fraudulent transactions tend to occur **earlier in the day** compared to legitimate transactions. The median transaction hour is **9:00 AM** for fraudulent transactions, compared to **12:00 PM** for legitimate transactions, while the average transaction hour is also lower (**10.38** vs. **11.56**). Although both legitimate and fraudulent transactions are distributed across the full 24-hour period, the lower average and median values indicate a noticeable tendency for fraudulent transactions to occur relatively earlier in the day.

3.5.7: Distance from Home by Fraud Status

The distributions of transaction distance from home are highly similar for legitimate and fraudulent transactions. Both groups have a median distance of **13.9 km**, while their average distances are nearly identical (**20.01 km** for legitimate transactions and **19.98 km** for fraudulent transactions). The considerable overlap between the distributions indicates that **the distance of a transaction from a customer's home does not significantly differentiate fraudulent transactions from legitimate ones**.

3.5.8: Time Since Last Transaction by Fraud Status

Fraudulent transactions tend to occur after a **slightly shorter interval** since the previous transaction compared to legitimate transactions. The median time since the last transaction is **7.80 hours** for fraudulent transactions and **8.35 hours** for legitimate transactions, while the average interval is also marginally lower (**11.56 hours** vs. **12.03 hours**). Despite this difference, the distributions overlap considerably, indicating that **time since the previous transaction alone is not a strong distinguishing factor for fraudulent activity**.

3.5.9: Number of Previous Transactions by Fraud Status

The distributions of previous transaction counts are nearly identical for legitimate and fraudulent transactions. Both groups have a median of **150 previous transactions**, with almost identical average values (**149.99** for legitimate transactions and **150.05** for fraudulent transactions). The substantial overlap between the distributions indicates that **customers' historical transaction counts do not significantly differentiate fraudulent transactions from legitimate ones**.

3.5.10: Monthly Transaction Frequency by Fraud Status

The distributions of monthly transaction frequency are highly similar for legitimate and fraudulent transactions. Both groups have a median of **20 transactions per month**, with nearly identical average frequencies (**20.00** for legitimate transactions and **20.01** for fraudulent transactions). The considerable overlap between the distributions indicates that **monthly transaction frequency alone is not a significant factor in distinguishing fraudulent transactions from legitimate ones**.

3.5.11: Fraud Rate by Merchant Category

The fraud rate varies considerably across different merchant categories, indicating that the type of merchant is associated with differing levels of fraud risk. **ATM Withdrawal** records the highest fraud rate (**8.74%**), followed closely by **Jewelry (8.70%)** and **Crypto Exchange (8.65%)**. In contrast, **Utilities (4.62%)**, **Grocery**, and **Entertainment** exhibit the lowest fraud rates. The difference between the highest- and lowest-risk categories is nearly twofold, suggesting that **merchant category is an important factor in understanding fraud patterns**.

3.5.12: Fraud Rate by Payment Method

Fraud is observed across all payment methods, indicating that no payment method is entirely immune to fraudulent activity. **Cheque** transactions exhibit the highest fraud rate (**5.67%**), followed closely by **Credit Card (5.60%)** and **Bank Transfer (5.51%)**, while **Debit Card** transactions record the lowest fraud rate (**5.46%**). The fraud rates across all payment methods fall within a narrow range, suggesting that **payment method alone has only a limited influence on the likelihood of fraudulent transactions**.

3.5.13: Fraud Rate by Device Type

Fraudulent transactions are observed across all device types, indicating that no transaction platform is exclusively associated with fraud. **ATM transactions** exhibit the highest fraud rate **(5.61%)**, followed closely by **Mobile (5.57%)** and **Desktop (5.47%)** devices, while **POS Terminal** records the lowest fraud rate **(5.44%)**. The fraud rates across all device types vary only slightly, suggesting that **device type alone is not a strong differentiating factor for fraudulent transactions**.

3.5.14: Fraud Rate by Day Type

Fraudulent transactions occur on both **weekdays** and **weekends**, indicating that fraud is not confined to a particular part of the week. **Weekend** transactions exhibit a slightly higher fraud rate (**5.57%**) compared to **weekday** transactions (**5.51%**); however, the difference is minimal. This suggests that **the likelihood of fraud remains relatively consistent throughout the week**, despite weekdays accounting for a higher overall transaction volume.

3.5.16: Fraud Rate by Time of Day

Fraudulent transactions occur during both **daytime** and **nighttime**; however, the fraud rate is considerably higher during nighttime hours. **Nighttime transactions** exhibit a fraud rate of **7.96%**, compared to **4.07%** for **daytime transactions**, making the nighttime fraud rate nearly twice as high. This indicates a **strong association between transaction timing and fraudulent activity**, suggesting that transactions performed at night are at a significantly higher risk of fraud.

3.5.17: Fraud Rate by Transaction Type (Domestic vs. International)
Fraudulent transactions occur in both **domestic** and **international** transactions; however, the fraud rate is significantly higher for international transactions. **International transactions** exhibit a fraud rate of **9.86%**, compared to **4.76%** for **domestic transactions**, making the fraud rate for international transactions more than twice as high. This indicates a **strong association between international transactions and fraudulent activity**, suggesting that cross-border transactions carry a substantially higher fraud risk.

3.5.18: Fraud Rate by Recent PIN Change Status
Fraudulent transactions occur among both customers who have and have not recently changed their PIN; however, the fraud rate is noticeably higher for customers with a recent PIN change. Transactions involving customers who **recently changed their PIN** exhibit a fraud rate of **8.36%**, compared to **5.28%** for those who did not. This difference suggests a potential association between recent PIN changes and fraudulent activity, indicating that such transactions may warrant additional attention during fraud monitoring.

3.5.19: Fraud Rate by Failed Authentication Attempts
The fraud rate increases substantially with the number of failed authentication attempts. Transactions with **0 or 1 failed attempt** exhibit relatively low fraud rates of approximately **4.5%**, whereas transactions with **2 or more failed attempts** show a sharp increase to around **14.6%**, with similarly high rates observed for **3, 4, and 5** failed attempts. This indicates a **strong association between repeated authentication failures and fraudulent activity**, making multiple failed-authentication attempts a significant indicator of potential fraud.

3.5.20: Fraud Type Distribution
The fraudulent transactions are classified into **six distinct fraud categories**, each representing a different type of fraudulent activity. The distribution is highly balanced, with each category contributing approximately **9,170 to 9,252** transactions. **Phishing** is the most common fraud type, while **Identity Theft** is the least common; however, the differences in their frequencies are minimal. This balanced representation ensures that **no single fraud category dominates the dataset**, allowing each type of fraud to be analyzed effectively.

3.5.21: Fraud Rate by Country
Fraudulent transactions are observed across all countries represented in the dataset, indicating that fraudulent activity is geographically widespread. **Brazil** records the highest fraud rate (**5.67%**), while **France** has the lowest (**5.42%**). The difference between the highest and lowest fraud rates is relatively small, suggesting **minimal geographical variation in fraud prevalence**. Despite considerable differences in transaction volumes across countries, the fraud rates remain consistently similar, indicating that **country alone is not a strong predictor of fraudulent activity**.

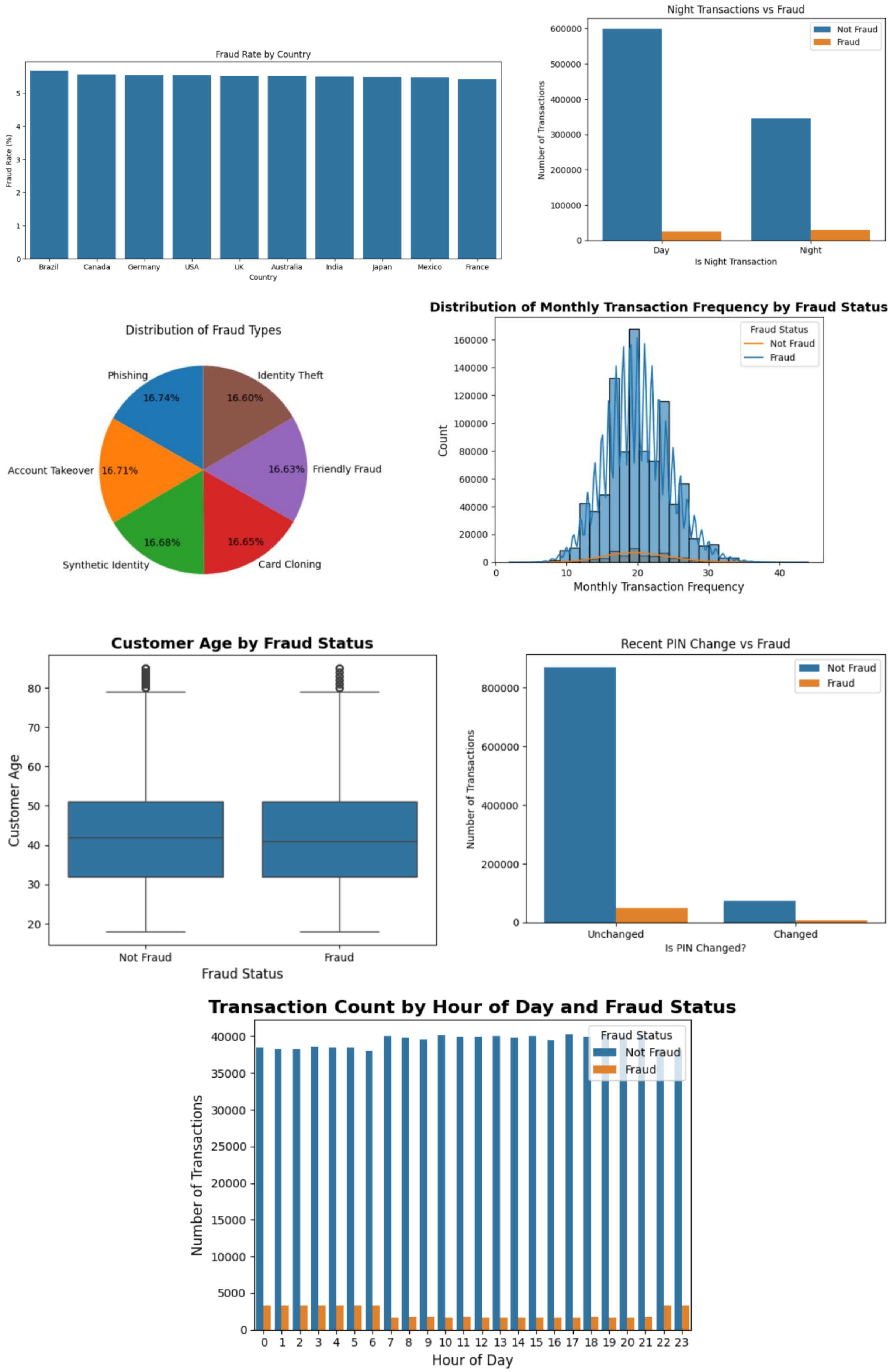


Figure 3.4: Few Bivariate Distributions

3.6: Correlation Analysis

Correlation Analysis was performed to measure the strength and direction of linear relationships among the numerical features in the dataset. Understanding these relationships helps identify variables that exhibit similar behaviour, detect potential multicollinearity, and determine which features are more closely associated with fraudulent transactions.
In this project, the Pearson correlation coefficient was used to evaluate pairwise correlations between numerical variables. The analysis was presented using a correlation matrix, a correlation heatmap, and a feature-wise correlation with the target variable (**is_fraud**). These visualizations provide a comprehensive overview of feature relationships and help identify variables that may contribute to fraud detection while revealing whether any strong linear dependencies exist within the dataset.

3.6.1: Correlation Matrix

A correlation matrix was generated to examine the linear relationships among all numerical features in the dataset. The matrix contains Pearson correlation coefficients ranging from **-1 to +1**, where values close to **+1 indicate a strong positive relationship**, values close to **-1 indicate a strong negative relationship**, and values near **0 indicate little or no linear relationship**.

3.6.1: Correlation Heatmap

A correlation heatmap was used to visualize the correlation matrix through a color-coded representation of the Pearson correlation coefficients. Different color intensities indicate the strength and direction of relationships between numerical features, enabling patterns to be identified more easily than by examining numerical values alone. The heatmap provides a quick visual summary of feature relationships, helping to identify highly correlated variables.

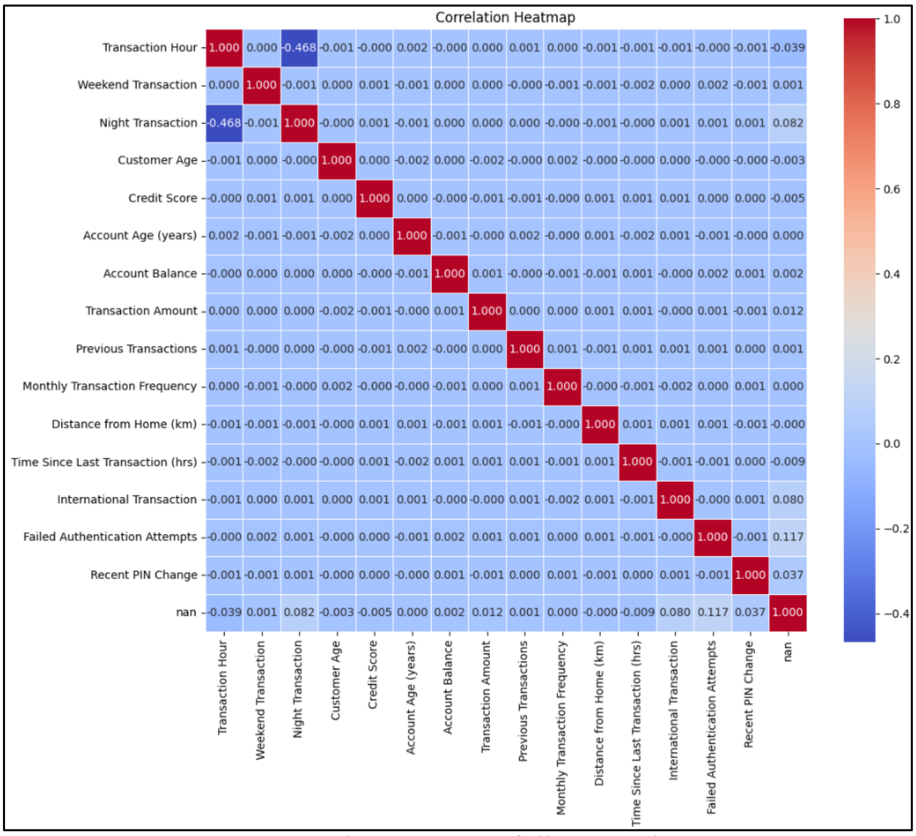


Figure 3.5: Correlation Heatmap of all Numerical Features

3.6.2: Correlation with Fraud

The correlation analysis indicates that **no numerical feature exhibits a strong linear relationship with fraudulent transactions**, as all correlation coefficients are relatively close to zero. Among the analyzed variables, **failed authentication attempts** show the strongest positive correlation with fraud (**0.117**), followed by **night transactions (0.082)** and **international transactions (0.080)**, suggesting that these security-related features are more closely associated with fraudulent activity than the remaining variables. Conversely, **hour of day** exhibits a weak negative correlation (**−0.039**), indicating a slight inverse relationship with fraud. Most financial and customer-related variables, including transaction amount, account balance, customer age, credit score, account age, and transaction frequency, display negligible correlations with the target variable. Overall, the findings suggest that **fraud is influenced by the interaction of multiple factors rather than any single numerical feature**, reinforcing the importance of multivariate analysis for uncovering more complex fraud patterns.

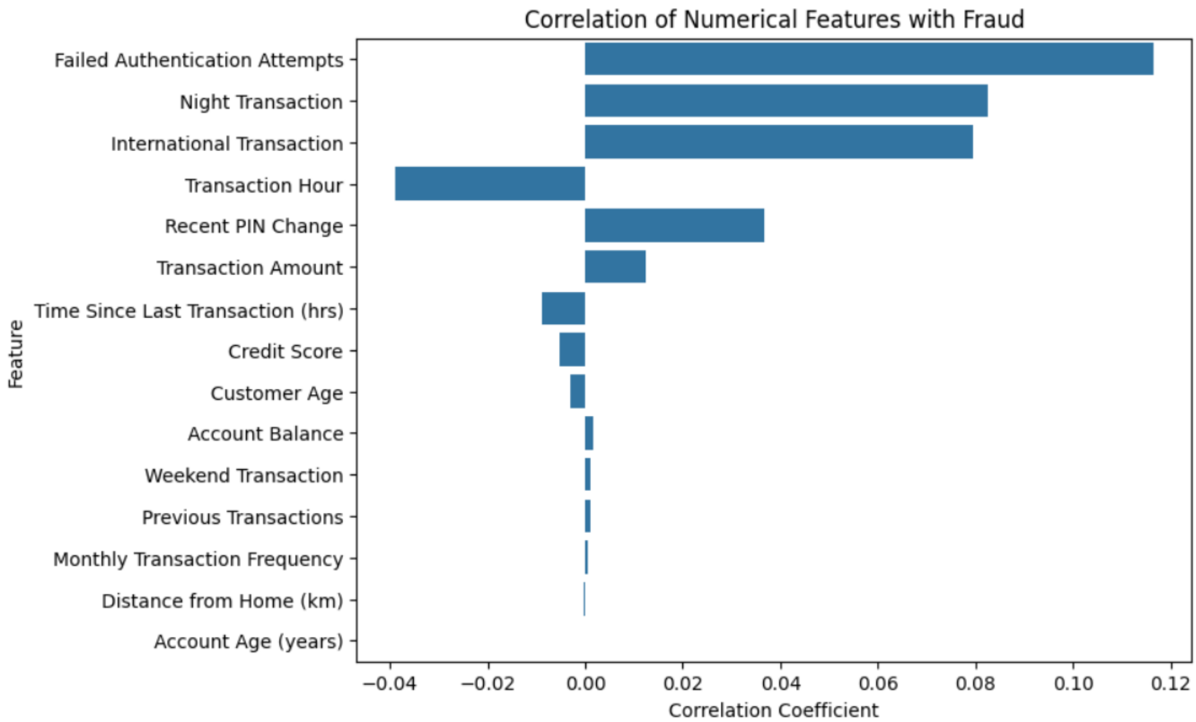


Figure 3.6: Correlation of numerical features with fraud

3.7: Multivariate Analysis

Multivariate Analysis was performed to examine the combined influence of multiple variables on fraudulent transactions. While univariate and bivariate analyses provide insights into individual features and pairwise relationships, multivariate analysis enables the identification of more complex patterns that emerge when several variables are considered simultaneously.
The objective of this analysis was to explore interactions among customer characteristics, transaction behaviour, security-related features, and geographical attributes to better understand the factors associated with fraudulent activity. By analysing multiple dimensions together, it becomes easier to identify high-risk scenarios, uncover hidden relationships, and derive insights that may not be apparent from individual variables alone.

3.7.1: Fraud Risk Overview Dashboard

The Fraud Risk Overview Dashboard highlights variations in fraud prevalence across multiple transaction characteristics. Among the analyzed features, **merchant category** exhibits the greatest variation, with **ATM Withdrawal (8.74%)**, **Jewelry (8.70%)**, and **Crypto Exchange (8.65%)** recording the highest fraud rates. In comparison, fraud rates across **payment methods**, **device types**, **countries**, and **cities** remain relatively consistent, with only minor differences between categories. The distribution of fraud types is also highly balanced, indicating that no single fraud category dominates the dataset. Overall, the findings suggest that **merchant category provides greater discriminatory value for fraud analysis than geographical, payment, or device-related characteristics**, making it a more informative feature for identifying high-risk transactions.

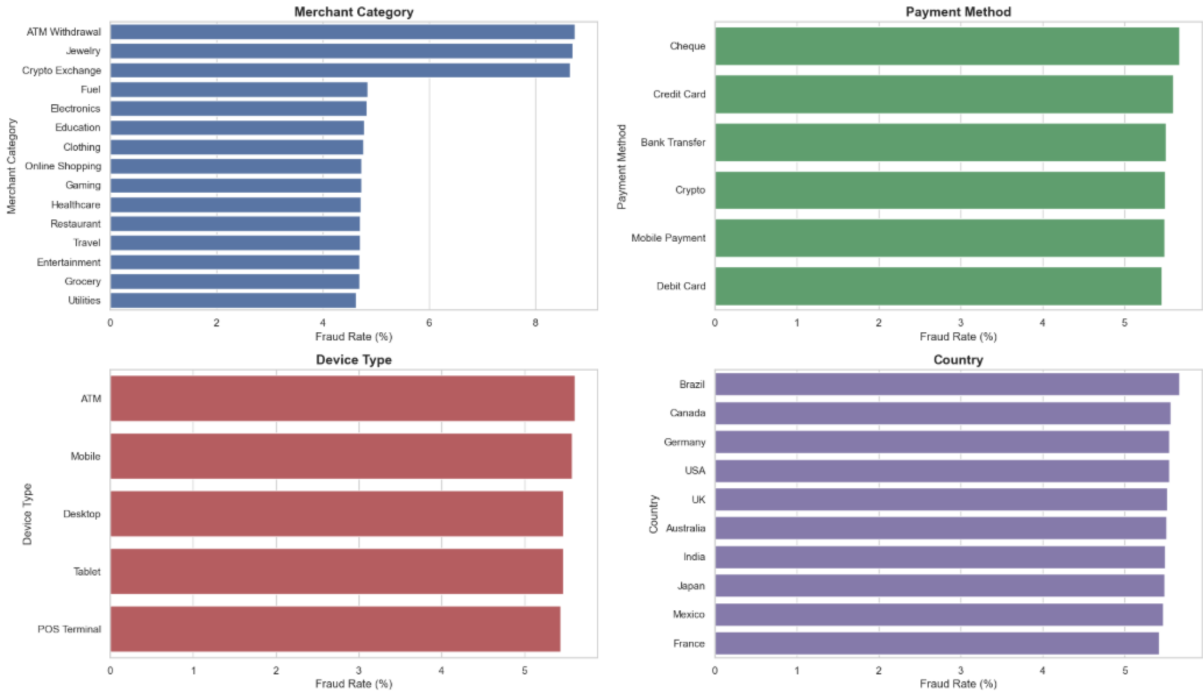


Figure 3.7: Fraud Risk Overview Dashboard

3.7.2: Security Features Dashboard

The Security Features Dashboard demonstrates that security-related variables exhibit the strongest association with fraudulent transactions. Fraud rates increase sharply after **two or more failed authentication attempts**, rising from approximately **4.5%** for zero or one failed attempt to around **14.5%** for two or more attempts. **International transactions** record the highest fraud rate (**9.86%**), followed by **nighttime transactions (7.96%)** and transactions involving **recent PIN changes (8.36%)**, all of which are substantially higher than their respective counterparts. These findings indicate that **security-related features provide significantly greater discriminatory power than most customer, financial, or geographical characteristics**, making them the most informative indicators for identifying high-risk transactions.

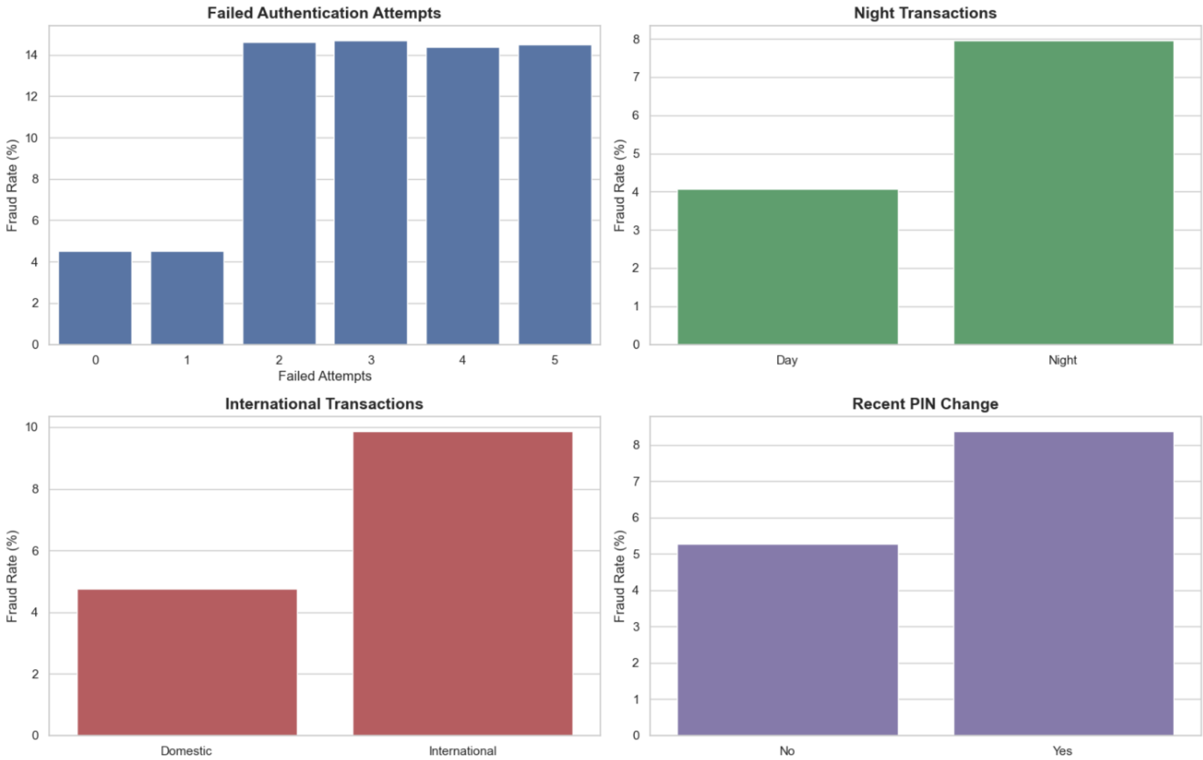


Figure 3.8: Security Features Dashboard

3.7.3: Customer and Financial Dashboard

The Customer & Financial Dashboard indicates that customer demographics and financial characteristics are broadly similar for legitimate and fraudulent transactions. Fraudulent transactions involve **slightly higher transaction amounts** and **marginally higher account balances** on average; however, both variables exhibit substantial overlap between the two classes. Customer-related features, including **customer age**, **credit score**, and **account age**, display nearly identical distributions, with only negligible differences in their statistical measures. Although transaction amount and account balance contain several high-value outliers, the overall findings suggest that **customer and financial attributes alone have limited ability to distinguish fraudulent transactions and are more effective when analyzed alongside stronger security-related features**.

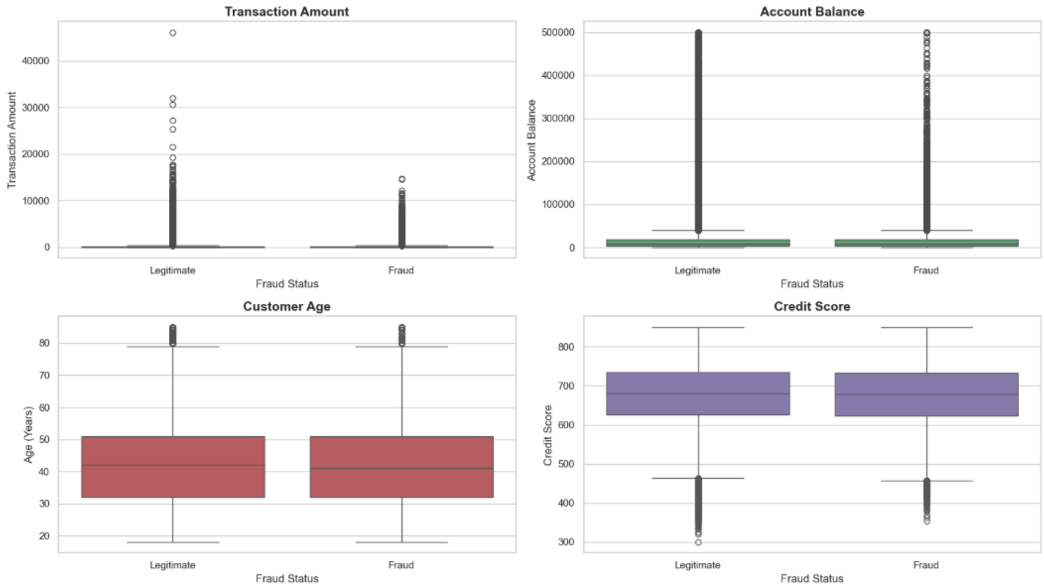


Figure 3.9: Customer and Financial Dashboard

3.7.4: Transaction Behaviour Dashboard

The Transaction Behaviour Dashboard shows that most behavioral features exhibit similar distributions for legitimate and fraudulent transactions. Fraudulent transactions tend to occur **slightly earlier in the day** and after a **slightly shorter interval since the previous transaction**; however, these differences are relatively modest. Features such as **distance from home**, **number of previous transactions**, and **monthly transaction frequency** display substantial overlap between the two fraud classes, with nearly identical statistical distributions. Overall, the findings indicate that **transaction behavior alone provides limited discriminatory power for fraud detection and is more informative when combined with security-related indicators**.

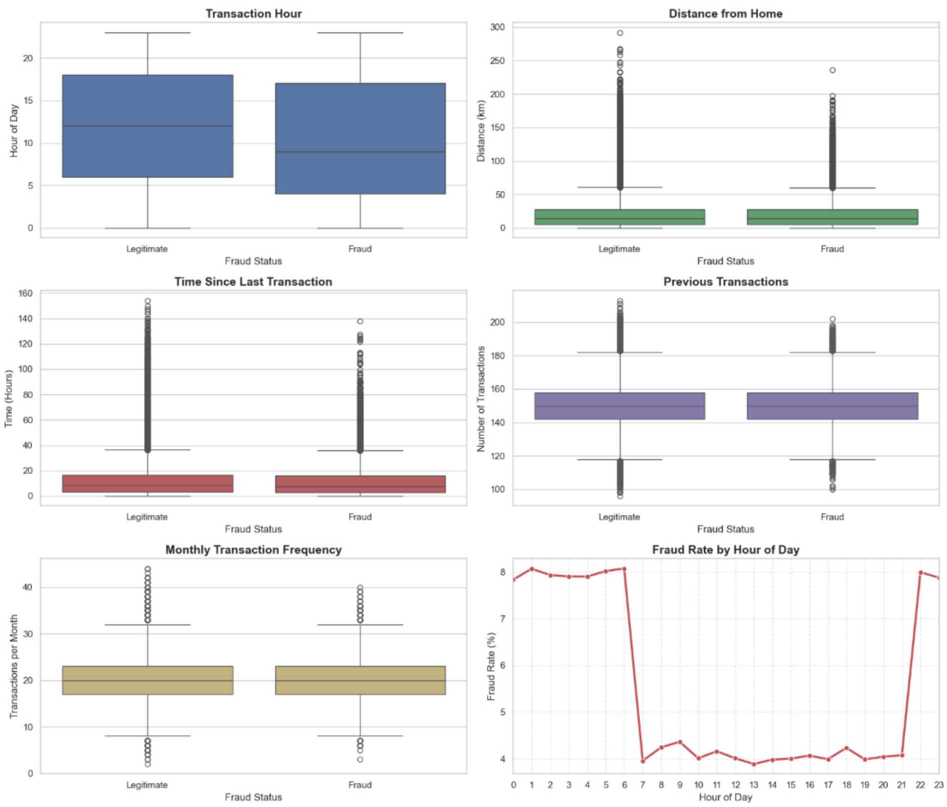


Figure 3.10: Transaction Behaviour Dashboard

3.7.4: Overall Fraud Risk Summary

The Overall Fraud Risk Summary consolidates the highest-risk categories identified across the exploratory analysis. Among all features, **two or more failed authentication attempts** exhibit the highest fraud rate (**14.69%**), followed by **international transactions (9.86%)**, **ATM Withdrawal merchant category (8.74%)**, **recent PIN changes (8.36%)**, and **nighttime transactions (7.96%)**. In comparison, the highest-risk **payment method**, **device type**, **country**, and **city** all record fraud rates of approximately **5.6%**, indicating relatively small differences across these categories. Overall, the analysis demonstrates that **security-related features consistently provide the strongest indicators of fraudulent activity**, while merchant category offers additional predictive value compared with geographical and other transaction characteristics.

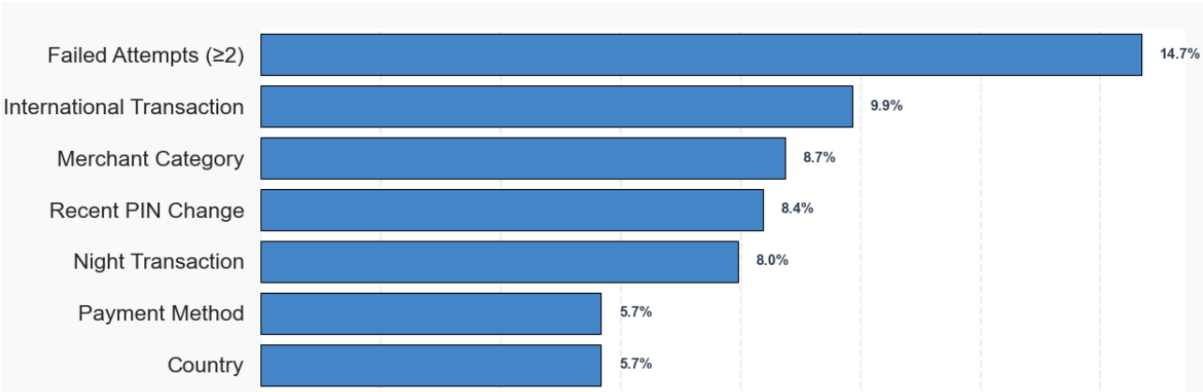


Figure 3.11: Overall Fraud Risk Summary

3.8: Dashboard Development

To complement the exploratory data analysis, an interactive web-based dashboard was developed using **Streamlit**. The dashboard was designed to present the analysis in an intuitive and user-friendly manner, enabling users to explore the dataset, visualize fraud patterns, and interact with key insights without requiring programming knowledge.

3.8.1: Dashboard Overview

The dashboard provides a user-friendly interface for exploring the dataset, analyzing fraud patterns, and viewing key insights through interactive visualizations. It enables users to navigate different sections of the project without requiring any programming knowledge, making the analytical results more accessible to both technical and non-technical users. The dashboard integrates the processed dataset with interactive charts and graphs created using **Plotly**, while **custom CSS** was used to improve the visual appearance and overall user experience. It is organized into multiple modules, including the project overview, dataset exploration, univariate analysis, bivariate analysis, correlation analysis, multivariate analysis, recommendations, and future scope. This modular design allows users to seamlessly explore different aspects of the analysis while maintaining a consistent and intuitive interface. Overall, the dashboard serves as an effective visualization and decision-support tool by transforming static analytical results into an interactive platform for exploring transaction fraud patterns and gaining actionable insights.

3.8.2: Dashboard Modules

The dashboard is organized into multiple modules that enable users to explore the project in a structured and interactive manner. The **Home** module provides an overview of the project along with key metrics and highlights. The **Dataset** module presents the processed dataset and its feature information. The **Univariate**, **Bivariate**, and **Multivariate Analysis** modules visualize the data from different analytical perspectives, allowing users to explore feature distributions, relationships, and complex fraud patterns. The **Final Insights** module summarizes the key findings obtained from the exploratory data analysis, while the **Security Recommendations** module presents actionable measures for mitigating fraud risks based on the analysis. The **Future Scope** module outlines potential enhancements and directions for extending the project in the future.

3.8.3: Dashboard Screenshots

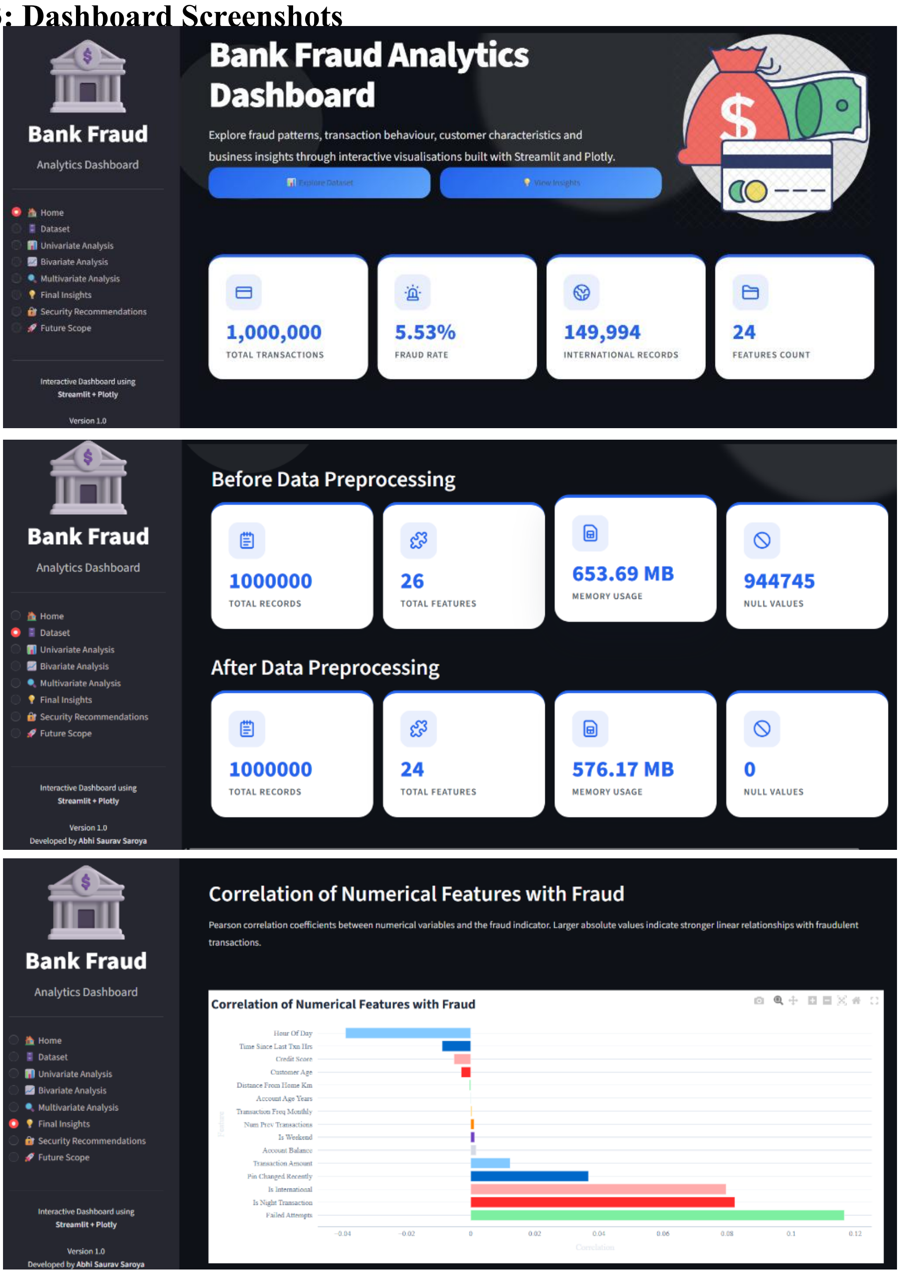


Figure 3.12: Dashboard Screenshots

3.9 Challenges Faced

During the development of the project, several challenges were encountered across data preprocessing, exploratory analysis, and dashboard implementation. Handling a dataset containing one million transaction records required memory optimization to ensure efficient processing and visualization. Data preprocessing involved cleaning the dataset, handling missing values appropriately, converting date and time fields into suitable formats, and removing unnecessary features to improve data quality.

Another challenge was selecting the most effective visualizations to communicate analytical insights while maintaining clarity and readability. Designing an interactive Streamlit dashboard also required careful organization of multiple analysis modules, integration of interactive Plotly charts, implementation of custom CSS styling, and optimization of application performance to provide a smooth user experience. Ensuring consistency between the notebook analysis and the dashboard visualizations, debugging implementation issues, and validating the analytical results were additional challenges that were successfully addressed throughout the project development process.

Chapter 4: **RESULTS AND CONCLUSIONS**

4.1: Key Findings

The major findings obtained from the exploratory data analysis are summarized below:

1. **Failed Authentication Attempts:** Transactions involving **two or more failed authentication attempts** exhibited the highest fraud rate (approximately **14.69%**), making repeated authentication failures the strongest indicator of fraudulent activity in the dataset.
2. **International Transactions:** International transactions recorded a fraud rate of **9.86%**, which is more than twice the fraud rate of domestic transactions (**4.76%**). This indicates that cross-border transactions are considerably more susceptible to fraud.
3. **Merchant Category:** Fraud prevalence varied significantly across merchant categories. **ATM Withdrawals (8.74%)**, **Jewelry (8.70%)**, and **Crypto Exchange (8.65%)** recorded the highest fraud rates, highlighting merchant category as one of the most informative transaction characteristics.
4. **Recent PIN Changes:** Transactions involving customers who had **recently changed their PIN** exhibited a higher fraud rate (**8.36%**) compared to customers without recent PIN changes (**5.28%**), suggesting that recent credential modifications are associated with elevated fraud risk.
5. **Time of Transaction (Hour of Day):** Fraudulent transactions tended to occur **earlier in the day** on average and were significantly more prevalent during **nighttime hours**, with nighttime transactions exhibiting a fraud rate of **7.96%** compared to **4.07%** during daytime, indicating that transaction timing plays an important role in fraud risk assessment.

4.2: Security Recommendations

Based on the findings of the exploratory data analysis, the following recommendations are proposed to strengthen fraud detection and prevention systems:

1. **Time-Based Fraud Detection:** Transactions performed during **nighttime hours** should be assigned higher risk scores, as they exhibit significantly higher fraud rates than daytime transactions. Additional verification measures, such as multi-factor authentication or real-time alerts, can be applied to high-value nighttime transactions.
2. **Failed Attempt Monitoring:** Transactions involving **two or more failed authentication attempts** should be immediately flagged for enhanced security checks. Automated monitoring systems can trigger temporary account restrictions, step-up authentication, or manual review to prevent unauthorized access and reduce fraud risk.
3. **Merchant Category Scoring:** Merchant categories with consistently higher fraud rates, particularly **ATM Withdrawals**, **Jewelry**, and **Crypto Exchange**, should be assigned higher fraud risk scores. Category-specific monitoring rules can improve fraud

detection accuracy while enabling financial institutions to prioritize transactions from high-risk merchants.

4. **Cross-Border Security:** Since **international transactions** exhibit substantially higher fraud rates than domestic transactions, additional security measures should be implemented for cross-border payments. These may include location verification, behavioral analysis, device validation, and adaptive authentication to ensure transaction legitimacy while minimizing inconvenience for genuine customers.

4.3: Future Scope

The project can be further enhanced by incorporating advanced analytical and intelligent fraud detection techniques. The following areas present significant opportunities for future development:

1. **ML-Based Fraud Detection:** The insights obtained from the exploratory data analysis can be utilized to train machine learning classification models such as **Random Forest**, **XGBoost**, and **Artificial Neural Networks** for automated fraud detection and improved predictive accuracy.

2. **Behavioral Analytics:** Additional behavioral features, including **customer spending patterns**, **transaction frequency**, **account age**, and **historical fraud records**, can be incorporated to better model customer behavior and enhance the detection of anomalous transactions.

3. **AI-Powered Risk Prediction:** An intelligent **risk-scoring system** can be developed to assign a fraud probability to every transaction by analyzing multiple transaction attributes simultaneously, enabling financial institutions to prioritize high-risk transactions for further investigation.

4. **Explainable Artificial Intelligence (XAI):** Explainable AI techniques can be integrated to provide transparent and interpretable explanations for why a transaction has been classified as fraudulent, thereby improving user trust, supporting regulatory compliance, and assisting investigators in decision-making.

5. **Real-Time Fraud Detection:** Streaming data analytics can be implemented to monitor transactions in real time, enabling suspicious activities to be detected instantly. Such a system would allow financial institutions to intervene before funds are transferred, significantly reducing financial losses caused by fraudulent transactions.

LEARNING OUTCOMES

The six-week industrial training and the development of the **Bank Fraud Exploratory Data Analysis (EDA)** project provided valuable practical exposure to real-world data analytics and fraud analysis. Throughout the project, I gained hands-on experience in collecting, preprocessing, cleaning, and analyzing large-scale transactional datasets using Python and its data science ecosystem.

The training strengthened my understanding of **NumPy**, **Pandas**, **Matplotlib**, **Seaborn**, **Plotly**, **Jupyter Notebook**, and **Streamlit**, enabling me to perform comprehensive exploratory data analysis and present insights through interactive visualizations. I also developed practical skills in data preprocessing, feature engineering, statistical analysis, correlation analysis, and interpreting analytical results to identify meaningful fraud patterns.

In addition, the project enhanced my ability to design and develop an interactive analytical dashboard, optimize data processing for large datasets, and effectively communicate complex findings through visualizations and reports. It also improved my problem-solving, debugging, and project management skills by exposing me to the complete lifecycle of a data analytics project—from data preparation and analysis to dashboard development and documentation.

Overall, this industrial training successfully bridged the gap between theoretical concepts and practical implementation, providing a strong foundation in data science, exploratory data analysis, and fraud analytics while preparing me for future projects involving machine learning and intelligent fraud detection systems.

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